



Term 3 PRT Individual Submission

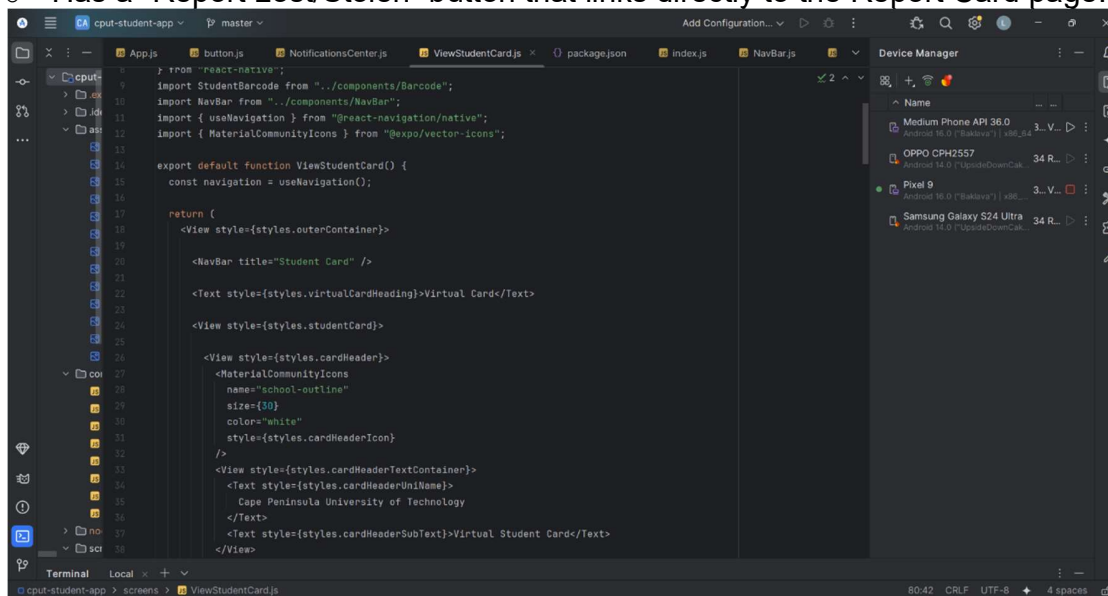
*01 September 2025
Project 362S*

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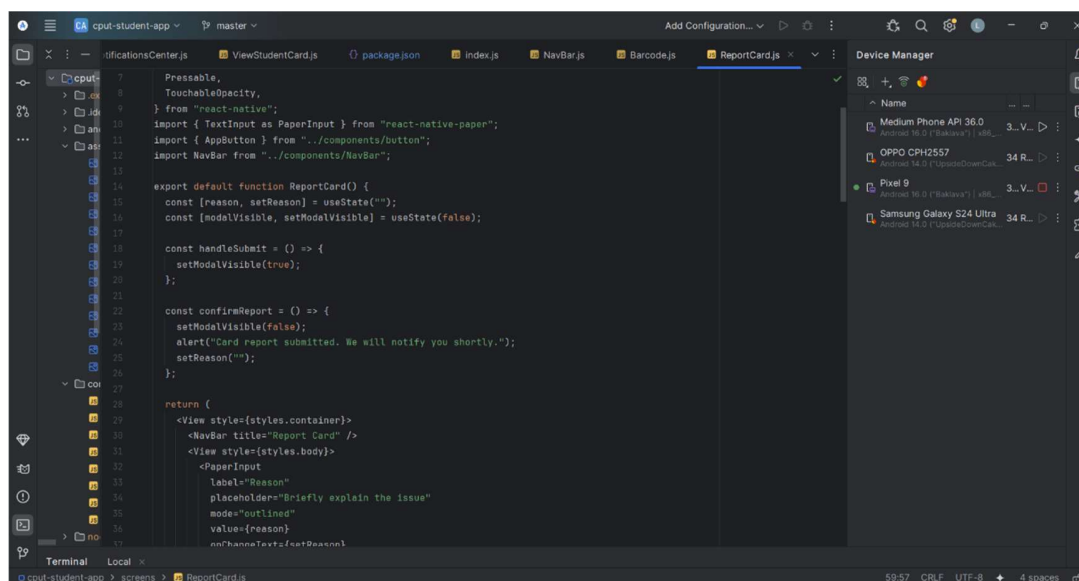
1. Coding Progress

I have developed multiple functional mobile screens for Android and iOS, and reusable components in React Native (Expo-managed workflow):

- Virtual Student Card Screen
 - Displays student details (name, number, faculty, email, etc.).
 - Includes a generated barcode for the student number.
 - Styled to resemble a real student ID card.
 - Has a “Report Lost/Stolen” button that links directly to the Report Card page.

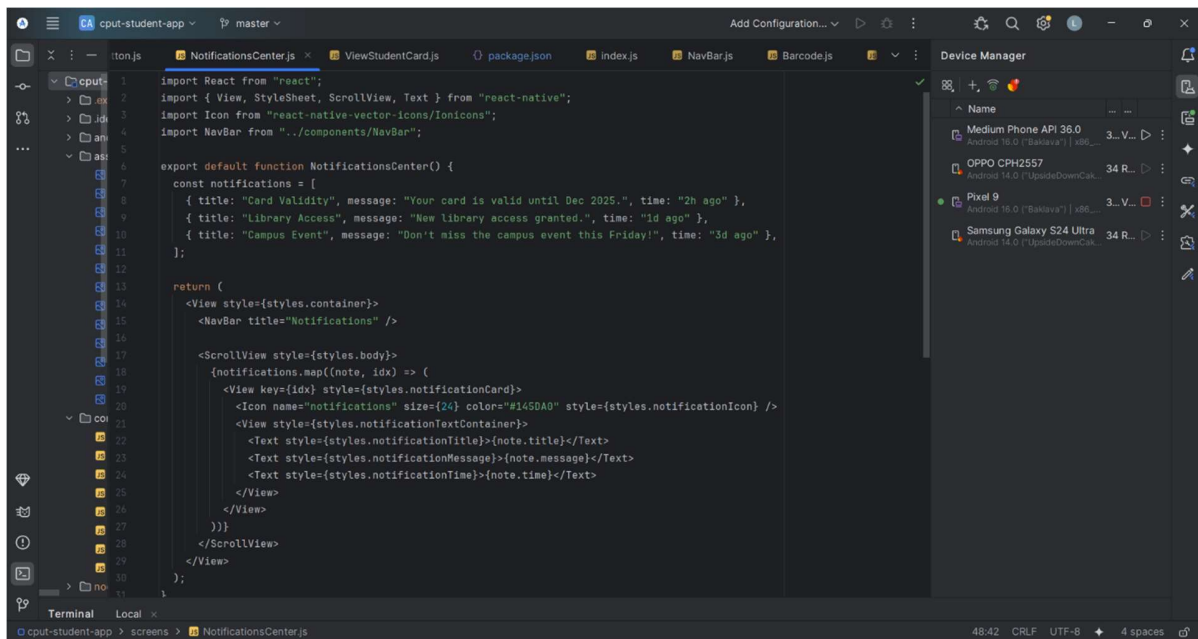


- Report Card Screen
 - Initially: free-text input + submit button.
 - Iterative improvements:
 - Added confirmation popup (system alert).
 - Replaced with custom modal styled to match the app.
 - Changed from free-text input to a dropdown list of predefined reasons (Spam, Fraud, Inappropriate, Other).
 - Disabled submit until a valid reason is chosen.



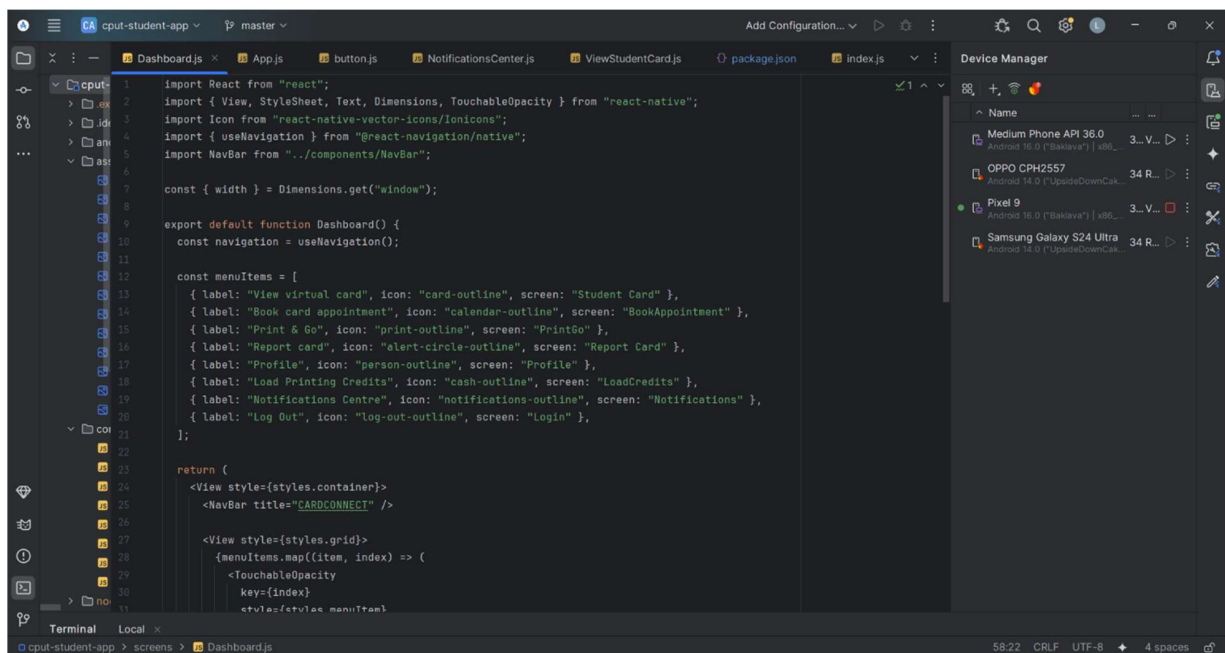
- Notifications Page

- Displays student notifications in a list of styled cards.
- Each notification has an icon, title, message, and timestamp.



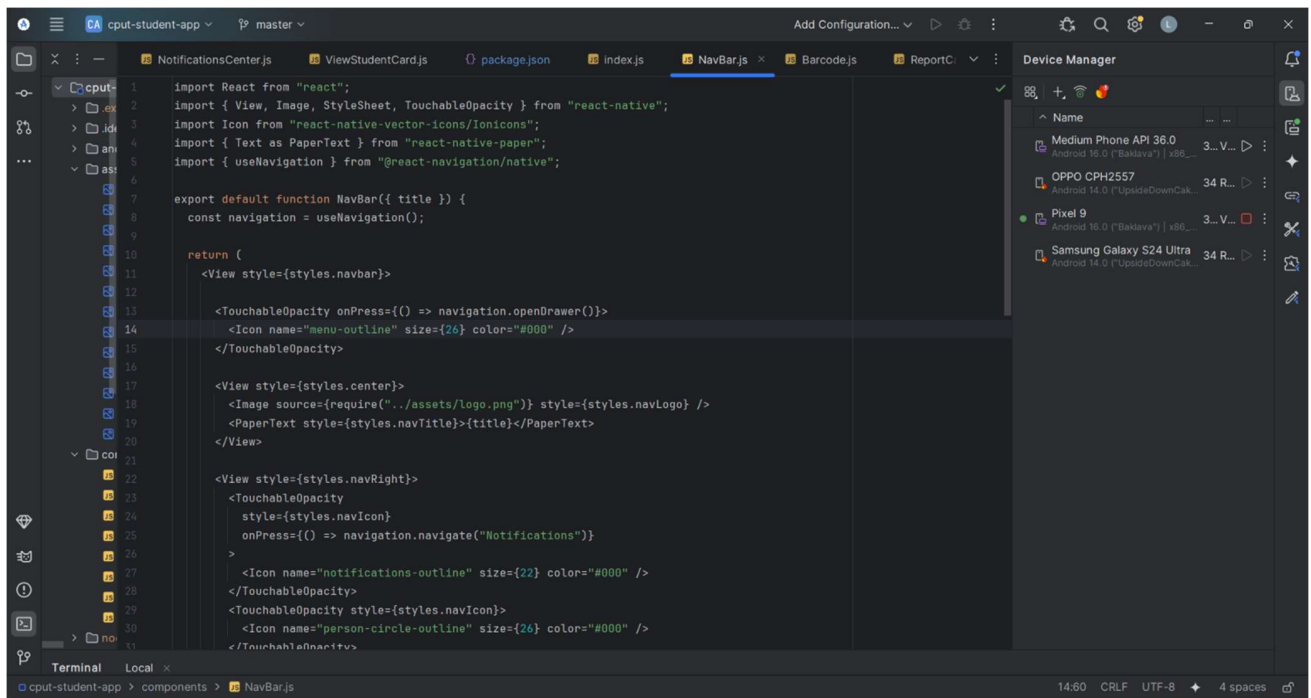
- Dashboard

- Central hub linking to all features:
 - View Virtual Card
 - Report Card
 - Notifications Centre
 - Load Printing Credits (planned)
 - Log Out
- Uses a grid menu layout for easy navigation.



- Reusable Components

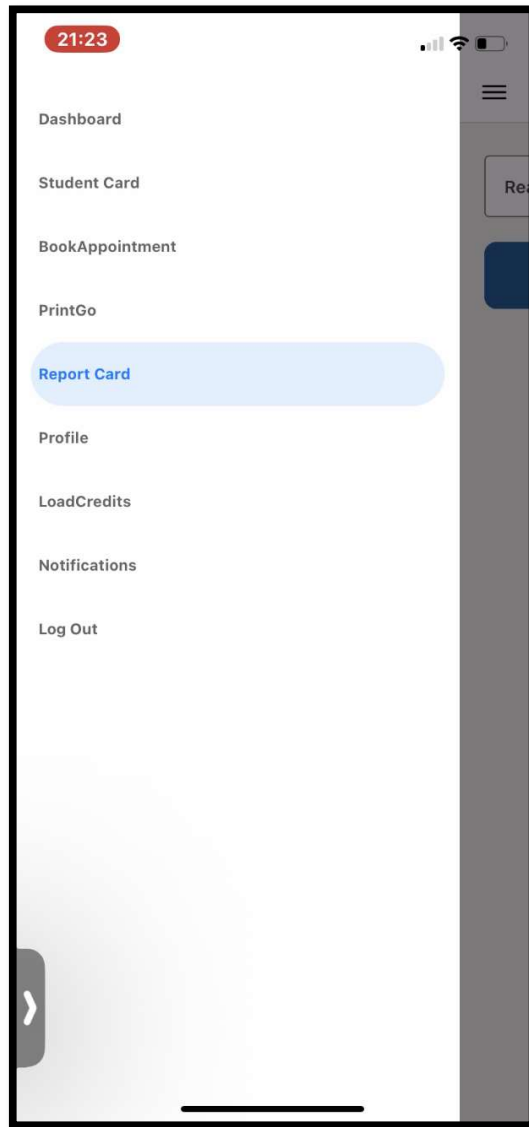
- NavBar – consistent across all screens, showing menu, title/logo, notifications, and profile icons.
- Burger Menu – collapsible side menu for quick access to app pages.



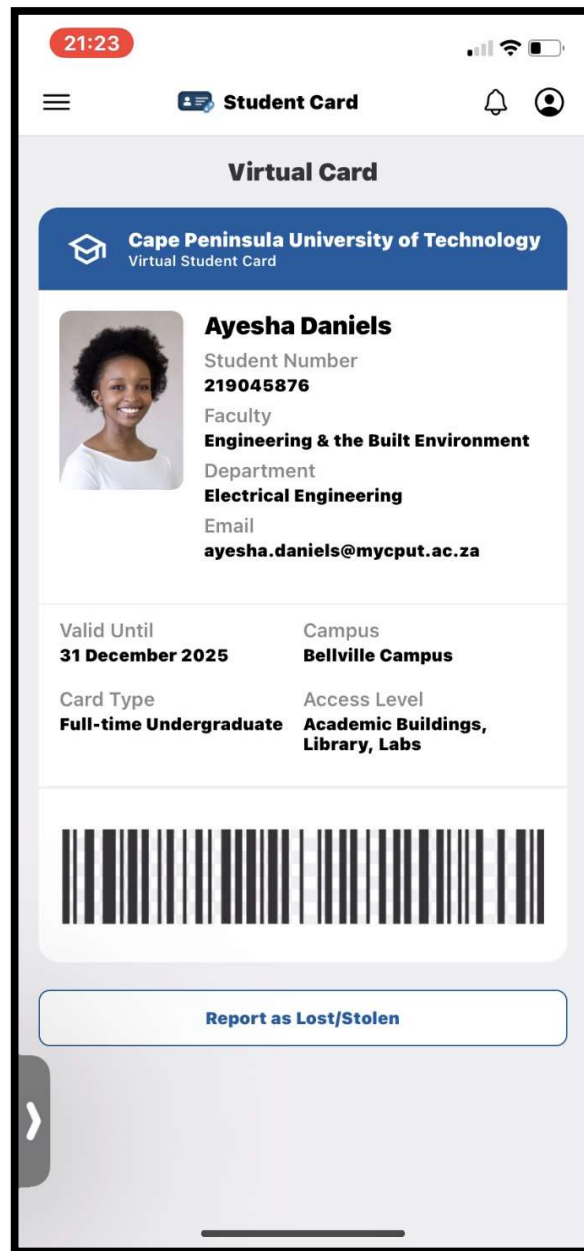
2. User Interface Design

The UI was improved step by step to create a cohesive, branded student experience:

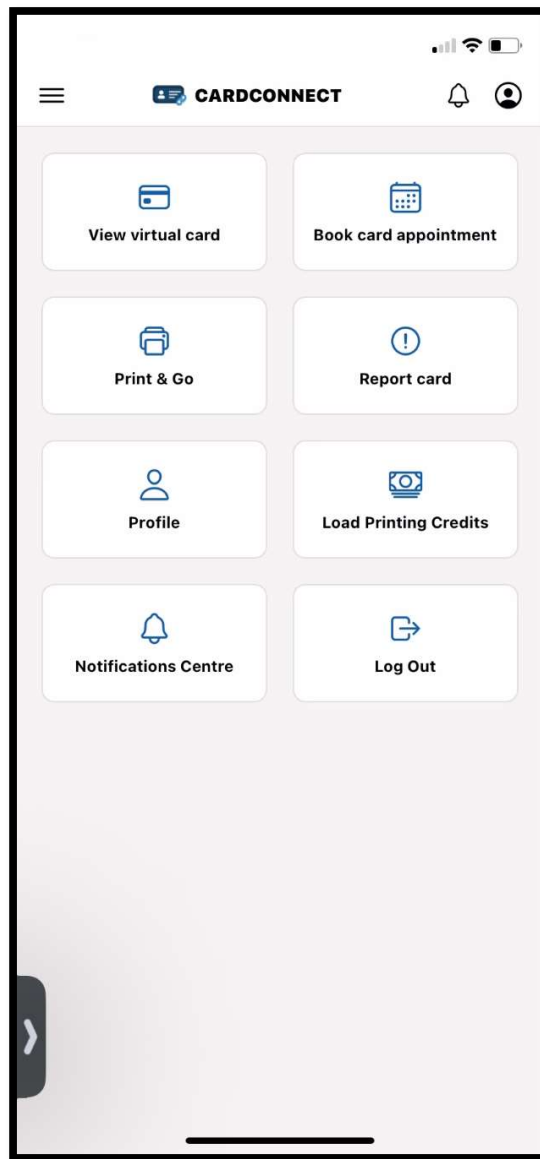
- Color Theme & Consistency
 - Primary blue: #145DA0
 - White and light-gray backgrounds
 - Rounded corners and shadows for cards/buttons



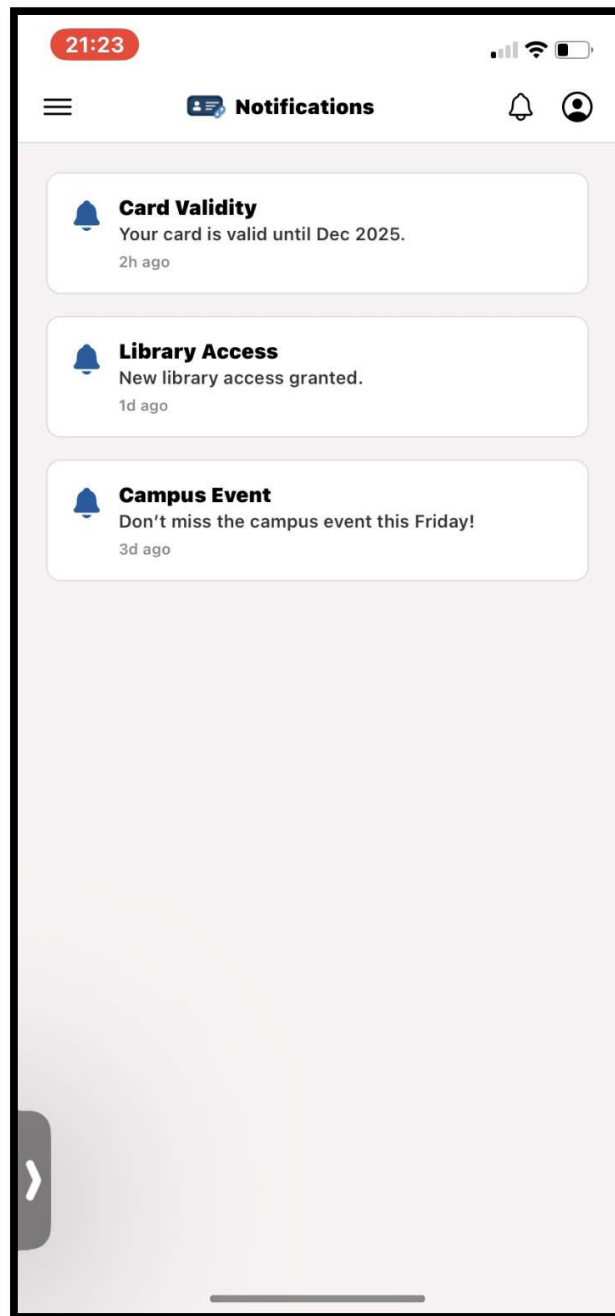
- Virtual Student Card UI
 - Clean ID card look with university logo and barcode.
 - Matches the professional feel of a real student card.



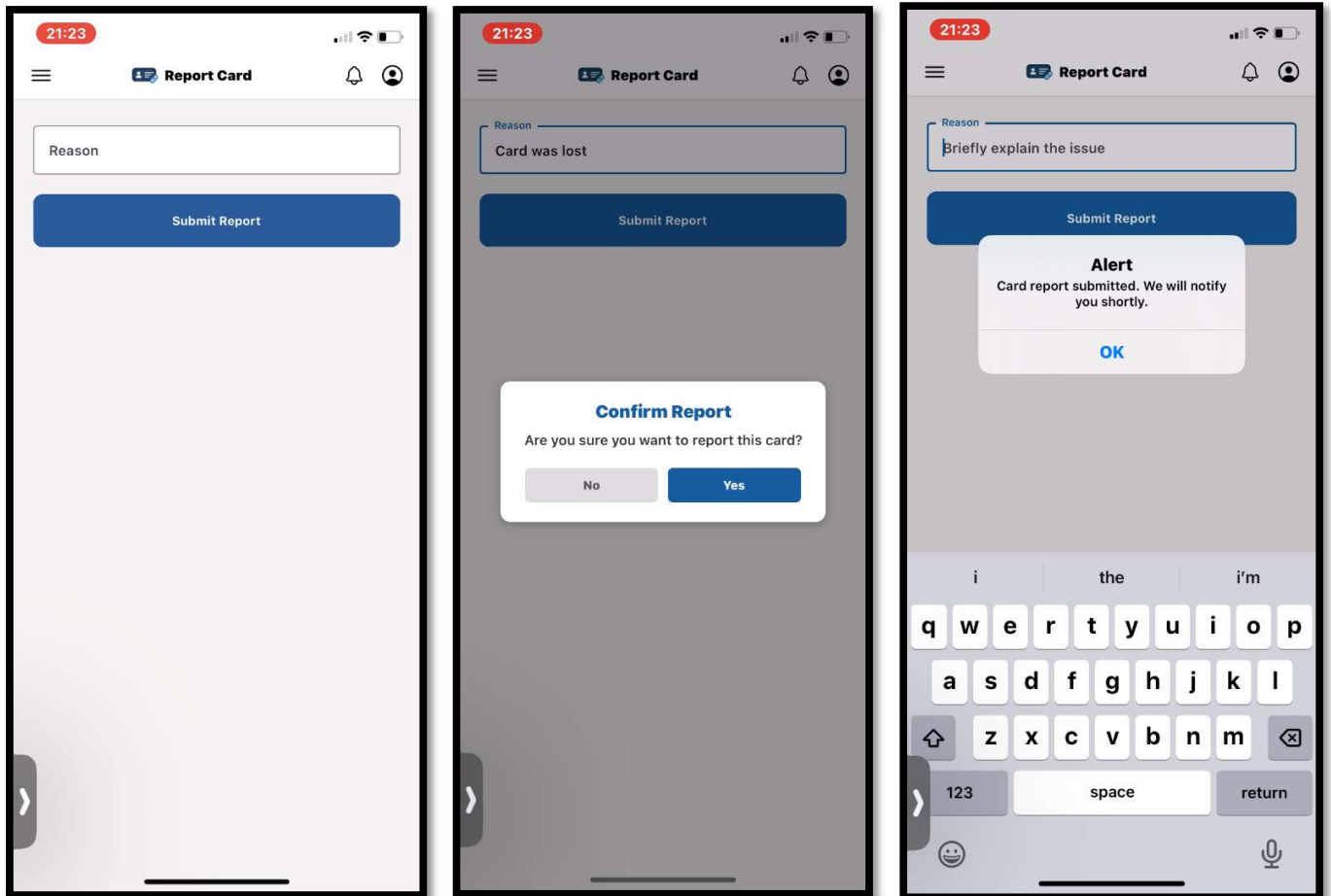
- Dashboard
 - Modern grid menu layout.
 - Icons + labels for quick navigation.



- Notifications
 - Card-based layout, styled for clarity and readability.



- Report Card Screen
 - Final design uses custom modal popup for confirmation.



Overall, the UI has evolved into a professional, unified design where all screens share the same nav bar and theme.

Must still be implemented:

- Report Card – Dropdown instead of text input.
- Dashboard – Less plain background.
- Virtual Card – QR code instead of barcode.

3. Data Design & Database Implementation (20%)

Data Requirements:

The system captures and manages the following data:

- Student Data
 - Student ID, Name, Faculty, Email, Barcode number.
- Reports
 - Report ID (Primary Key)
 - Student ID (Foreign Key)

- Reason (dropdown choice)
 - Timestamp
 - Status (Pending/Resolved).
- Notifications
 - Notification ID
 - Title
 - Message
 - Timestamp
 - Student ID (optional, if targeted).

4. Testing Evidence

Manual Test Cases

Feature	Input	Expected Output	Actual Output	Status
Android	Run on Android	Run on Android	Works	Pass
Report Card – no reason	No dropdown selection	Submit disabled	Button disabled	Pass
Report Card – valid reason	Select “Spam” → Submit	Modal asks to confirm	Modal appears	Pass
Confirm modal → No	Click No	Modal closes, no submission	Works	Pass
Confirm modal → Yes	Click Yes	Report submitted, input cleared	Works	Pass
Virtual Card	Open screen	Student details + barcode visible	Correct	Pass
Notifications	Open screen	Notifications list displayed	Correct	Pass
Dashboard navigation	Tap “View Virtual Card”	Redirect to card screen	Works	Pass
iOS	Run on iOS	Run on iOS	Works	Pass

Debugging Fixes

- Solved ARTShape crash by replacing react-native-barcode-svg with react-native-svg solution.
- Fixed asset import errors (missing images/logos).
- Standardized components (NavBar, Burger Menu) to avoid duplication.

The app has been tested at every stage to make sure navigation, UI, and reporting logic all work as intended.

Overall Progress Summary

- Coding: Multiple screens built (Dashboard, Virtual Student Card with barcode, Notifications, Report Card) plus reusable components (NavBar, Burger Menu).
- UI: Refined into a consistent, modern design with branding, custom modals, and dropdowns.
- Data: Structured schema for students, reports, and notifications. Database designed to 3NF.
- Testing: Test cases confirm correct functionality. Debugging resolved major crashes and inconsistencies.

5. Challenges Faced

- Implementing the barcode was pretty tricky and was a massive challenge. I used YouTube tutorials and websites like dev.to and nmpjs.com to help me out but in the end decided a QR code would look neater and be more modern for the Virtual Student Card.

- Learning to code in React and Android Studio was also something new to me, so it took time for me to adjust.
- I also had to learn how I could run the app using Expo Go in order to see it on the emulator and my iPhone.